

THE NH₃ HYPERFINE INTENSITY ANOMALY IN HIGH-MASS STAR-FORMING REGIONSMATTHEW A. CAMARATA¹, JAMES M. JACKSON¹, AND EDWARD CHAMBERS^{1,2}¹ Institute for Astrophysical Research, Boston University, Boston, MA 02215, USA; camarata@bu.edu, jackson@bu.edu² Physikalisches Institut, Universität zu Köln, D-50937 Köln, Germany; chambers@ph1.uni-koeln.de

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ABSTRACT

Anomalous ammonia (NH₃) spectra, exhibiting asymmetric hyperfine satellite intensity profiles in the $(J, K) = (1, 1)$ inversion transition, have been observed in star-forming regions for over 35 years. We present a systematic study of this “hyperfine intensity anomaly” (HIA) toward a sample of 334 high-mass star forming regions: 310 high-mass ($\gtrsim 100 M_{\odot}$) clumps and 24 infrared dark clouds. The HIA is ubiquitous in high-mass star forming regions. Although LTE excitation predicts that the intensity ratios of the outer satellites and inner satellites are exactly unity, for this sample the ensemble average ratios are 0.812 ± 0.004 and 1.125 ± 0.005 , respectively. We have quantified the HIA and find no significant relationships between the HIA and temperature, line width, optical depth, and the stage of stellar evolution. The fact that HIAs are common in high-mass star-forming regions suggests that the conditions that lead to HIAs are ubiquitous in these regions. A possible link between HIAs and the predictions of the competitive accretion model of high-mass star formation is suggested; however, the expected trends of HIA strength with clump evolutionary stage, rotational temperature, and line width for evolving cores in competitive accretion models are not found. Thus, the exact gas structures that produce HIAs remain unknown. Turbulent gas structures are a possible explanation, but the details need to be explored.

Key words: ISM: clouds – ISM: molecules – stars: formation

Supporting material: machine-readable table

1. INTRODUCTION

Ammonia (NH₃) inversion transitions are split into five hyperfine line components due to nuclear quadrupole interactions. This hyperfine splitting results in a brighter central “main line” flanked on either side by two pairs of symmetrically placed satellite lines. In LTE, the outer satellites and inner satellites are each predicted to have equal intensities. However, anomalous NH₃ inversion spectra exhibiting deviations from the LTE-predicted hyperfine intensities have been observed in a number of sources over the past 35 years. The anomaly itself, hereafter referred to as the “hyperfine intensity anomaly” (HIA), is characterized as an asymmetry in the intensities of the inner and outer satellite pairs of the $(J, K) = (1, 1)$ inversion transition. The HIA was first detected (in absorption) toward DR 21 by Matsakis et al. (1977). Subsequent observations demonstrated HIAs to be a common feature in many, though not all (Takano et al. 1986; Ungerechts et al. 1986), star-forming regions and warm molecular clouds, including S106, OMC2, W48 (see Stutzki et al. 1984 and references therein), M17 SW (Guesten & Fiebig 1988), W3(OH), and G34.3+02 (Keto et al. 1987). Recent studies provide additional support for the hypothesis that the HIA is common in star-forming regions (Caproni et al. 2002; Lopes et al. 2002; Fontani et al. 2012; Nishitani et al. 2012). Significant detections of HIAs have also been made toward multiple hot molecular cores (Longmore et al. 2007), compact regions of hot (~ 100 K) and dense ($\sim 10^5$ – 10^8 cm⁻³) molecular gas found near newly formed high-mass protostars (Garay & Lizano 1999; Kurtz et al. 2000; Churchwell 2002).

Matsakis et al. (1977) first proposed an explanation for the HIA by invoking selective radiative trapping in the non-metastable NH₃ $(J, K) = (2, 1) \rightarrow (1, 1)$ rotational transition (see also Stutzki & Winnewisser 1985). This mechanism requires the gas to have line widths of ~ 0.3 – 0.6 km s⁻¹ so that

overlap between the inner satellites and outer satellites is diminished, while the inner satellites and main line maintain significant overlap. Under these conditions, the optical depths of the $(2, 1) \rightarrow (1, 1)$ outer satellites are lowered compared to the inner satellites and main line. This lower optical depth leads to an increased escape probability for the outer-satellite photons and therefore decreased $(2, 1)$ level populations in certain hyperfine states. Because the outer satellites of the $(2, 1) \rightarrow (1, 1)$ rotational transition populate the $F = 0$ and $F = 1$ levels of the $(1, 1)$ metastable state, these $(1, 1)$ levels in turn become overpopulated. Given the allowed $\Delta F = 1$ inversion transitions, and the fact that the $F = 0$ state is more strongly overpopulated than the $F = 1$ state, the result is an increase in the intensity of the $F = 0 \rightarrow 1$ satellite (red side) and a decrease in intensity of the $F = 1 \rightarrow 0$ satellite (blue side). The reverse occurs for the inner satellites, with the $F = 1 \rightarrow 2$ satellite having a greater intensity than the $F = 2 \rightarrow 1$ satellite. The effect is diminished for the inner satellites, however, due to the remaining overlap between (and shuffling of photons among) the inner satellites and main line of the $(2, 1) \rightarrow (1, 1)$ transition (see Figure 1 for an example of an anomalous spectrum and Stutzki & Winnewisser 1985 for a more detailed discussion).

The selective trapping mechanism requires narrow line widths of ~ 0.3 km s⁻¹. However, because typical NH₃ spectra of molecular clouds have line widths on the order of 1–3 km s⁻¹, Matsakis et al. (1977) simultaneously adopted a model incorporating gas clumping to account for this disparity. In this HIA “core” model, the observed spectra are produced from a number of small cores possessing line widths of ~ 0.3 km s⁻¹, but with an overall dispersion among the cores of ~ 1 – 3 km s⁻¹. Subsequent numerical radiative transfer calculations by Stutzki & Winnewisser (1985) confirmed the validity of this model and describe the core properties necessary to produce the anomaly, namely diameters of $\sim 10^{-2}$ pc, densities

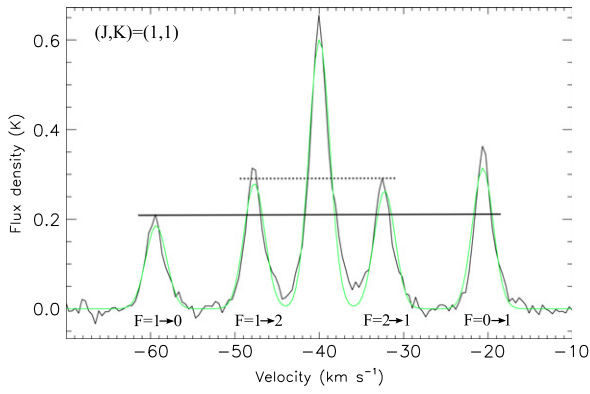


Figure 1. NH_3 $(J, K) = (1, 1)$ inversion spectrum toward G316.73+00.07 exhibiting a significant HIA. The $F = 1 \rightarrow 0$ satellite has a lower intensity than the $F = 0 \rightarrow 1$ satellite, while the $F = 1 \rightarrow 2$ satellite has a higher intensity than the $F = 2 \rightarrow 1$ satellite. The black line is the spectrum data and the green line is the Gaussian fit. The gray solid and dashed lines are fiducial markers for the outer and inner satellites respectively.

of $\sim 10^6$ – 10^7 cm^{-3} , and masses on the order of $1.0 M_\odot$. (In this paper we use the term “clump” to refer to larger (>0.1 pc), high-mass ($\gtrsim 100 M_\odot$) molecular regions that fragment into multiple cores and “core” to refer to smaller (<0.1 pc) condensations that form individual stars or binary pairs.)

In this paper, we extend the study of the HIA to a statistical analysis of a sample of 334 high-mass star-forming regions, with a separate analysis of a subsample of 24 infrared dark clouds (IRDCs). IRDCs are cold (< 25 K), massive ($>100 M_\odot$) molecular clouds thought to be the birthplaces of high-mass OB stars and star clusters (Rathborne et al. 2006, 2010). We seek to understand whether the HIA is a prevalent feature among high-mass star-forming regions, since its ubiquity may imply a common underlying condition or mechanism in the process of high-mass star formation, a still poorly understood phenomenon. Further, to probe the validity of the HIA core model, we compare the HIA to properties of high-mass star-forming regions expected to trace evolutionary sequence.

2. OBSERVATIONS AND ANALYSIS

We obtained data from two datasets, which together consist of a total of 981 sources. We obtained publicly released Effelsberg Telescope data from Wienen et al. (2012), who observed NH_3 $(1, 1)$, $(2, 2)$, and $(3, 3)$ lines toward 862 massive clumps ($\gtrsim 100 M_\odot$) detected by the APEX Telescope Large Area Survey of the GALaxy (ATLASGAL). These observations were carried out between 2008 and 2010 with the Effelsberg 100 m telescope. Observations were made using the AK90 and FFTS spectrometers in frequency-switching mode with spectral resolutions of 0.5 and ~ 0.7 km s^{-1} . The median system temperature was ~ 70 K. The total integration time per source was ~ 5 minutes. The beam size at 24 GHz is $40''$ (see Wienen et al. 2012 for details).

Additionally, we observed the NH_3 $(1, 1)$, $(2, 2)$, and $(3, 3)$ lines in 119 IRDCs with Galactic longitudes between 290° and 355° and latitudes between -1° and $+1^\circ$. These observations were carried out at the 64 m Parkes radio telescope in New South Wales, Australia³ from 2008 June 19–20 and 2009 June

12–13. Coordinates for each target were selected using IRDC candidates identified by Simon et al. (2006) from $8 \mu\text{m}$ *Midcourse Space Experiment* images. Data were taken using the K-band receiver and Parkes Digital Filter Bank (DFP3) backend in Spectrometer Mode at a frequency of 23.782 GHz, with a bandwidth of 256 MHz across 8192 channels. This yields a spectral resolution of 31.3 kHz (0.4 km s^{-1}) per channel with a beam size of $\sim 50''$. We observed each source in position-switching mode with a single pointing and typical integration time of 4.2 minutes on-source. To increase signal-to-noise, both polarizations were averaged. The typical noise was $T_{\text{rms}} \sim 0.01$ K. We reduced the data using standard methods with the ATNF Spectral Analysis Package.

Spectral line fitting was performed using our own custom IDL routines. Each spectrum was fit using five Gaussian functions, with fixed frequency separations corresponding to the known separations of the five hyperfine components. The line width was assumed to be identical for all hyperfine lines. This enabled us to obtain independent intensities for all components simultaneously. Spectral parameters and associated 1σ uncertainties were derived from fits using the non-linear least squares fitting routine MPFITFUN (Markwardt 2009). Along with a threshold signal-to-noise cut, we also used MPFITFUN’s reduced chi-square (χ^2) statistic to exclude poorly fitted sources from further analysis. After an initial fit, only lines having a signal-to-noise ratio ≥ 3.0 and a $\chi^2 < 4.0$, were considered for further analysis. However, if all components of the $(1, 1)$ transition did meet these criteria, but the $(2, 2)$ transition did not, the source was included in HIA calculations, but its rotational temperature and optical depth were not calculated. Each spectrum was subsequently checked by eye to ensure that it did not contain fitting abnormalities. Due to the potential for a non-anomalous source with multiple velocity components to mimic the HIA, such spectra were also excluded from further analysis. After applying these signal-to-noise and χ^2 thresholds, a total of 310 sources in the Effelsberg sample and 24 sources in the Parkes sample remained.

We quantified the HIA by considering separately the ratio of the line amplitudes of the outer and inner pairs of satellites of the NH_3 $(1, 1)$ transition. In LTE, where symmetric outer- and inner-satellite line intensities are expected, this ratio is exactly unity. Anomalous spectra that deviate from this LTE symmetry will thus have ratios that deviate from unity. Therefore, for each source, we calculated this “HIA ratio” for both the inner- and outer-satellite line pairs by taking the ratio of their respective fitted line amplitudes. First, we defined the outer satellite transitions $F = 0 \rightarrow 1$ as “right-outer” (RO) and $F = 1 \rightarrow 0$ as “left-outer” (LO). Similarly, we defined the inner satellite transitions $F = 1 \rightarrow 2$ as “left-inner” (LI) and $F = 2 \rightarrow 1$ as “right-inner” (RI). For an intensity (line amplitude) I , we then calculated the outer HIA ratio as

$$\text{HIA}_{\text{outer}} = \frac{I_{\text{LO}}}{I_{\text{RO}}}$$

and the inner HIA ratio as

$$\text{HIA}_{\text{inner}} = \frac{I_{\text{LI}}}{I_{\text{RI}}}.$$

We also calculated the inner and outer HIA ratio averages for the ensemble of sources in each sample using the arithmetic mean of the individual source HIAs; we did this for the full

³ The Parkes radio telescope is part of the Australia Telescope National Facility which is funded by the Commonwealth of Australia for operation as a National Facility managed by CSIRO.

sample of 334 sources and separately for the subsample of 24 IRDCs. HIA ratio uncertainties were propagated directly from the uncertainties in the fit parameters returned by MPFITFUN.

Calculations of NH_3 (1, 1) main line optical depths and NH_3 rotational temperatures were made using the canonical equations of Ho & Townes (1983). For this portion of the analysis we used a multi-Gaussian fit with fixed line amplitudes, since the equation describing optical depth relies on analytically derived LTE ratios of the (1, 1) inner and outer satellites to the main line. That is, the amplitudes of the outer satellites were set equal, and the amplitudes of the inner satellites were set equal. All lines were again assumed to have equal line widths. Optical depths were derived from the average of the ratios of the main component to the inner and outer satellites. Because the equation describing optical depth cannot be solved analytically, optical depth uncertainties were determined using a Monte Carlo analysis. This was accomplished by first generating Gaussian-distributed uncertainties corresponding to the 1σ uncertainties in each of the (1, 1) fitted line amplitudes. For each of $n = 10^5$ trials, the generated uncertainties were added to each fitted line amplitude and the optical depth was calculated. The optical depth uncertainty was then determined using the standard deviation of this distribution of generated optical depths. This method of uncertainty estimation was also used in the calculation of rotational temperature uncertainties by varying the (1, 1) and (2, 2) line amplitude uncertainties as well as the above calculated optical depth uncertainty.

Finally, in order to determine whether the “strength” (how far from unity the HIA ratio deviates) of the HIA varies as a function of the stage of high-mass stellar evolution, we classified each source by eye as either “quiescent,” “protostellar,” or “H II region” based upon the evolutionary classification scheme of Chambers et al. (2009). Using *Spitzer* IR images, we classified sources as quiescent if they showed no emission in either IRAC 3–8 μm or MIPS 24 μm , protostellar if a 24 μm point source or a “green fuzzy” at 4.5 μm was seen, and H II region if there was bright extended emission at 8 μm . If the source was unable to be classified, or was indeterminate, it was labeled as “unclassified.”

To verify the classification scheme for the “H II region” category, we searched the CORNISH VLA 5 GHz continuum survey for radio free-free emission at the position of the targets classified as H II regions based on the mid-IR *Spitzer* images. We searched for cataloged 5 GHz CORNISH sources near the target positions. Of the sources classified as H II regions in the high signal-to-noise subsample, 25 lie within the CORNISH survey region. Of these 25 sources, 11 have a CORNISH source within 0.6 arcmin of the target position, and 2 have a CORNISH source within 3 arcmin of the target position. There were no catalog entries within 3 arcmin for 12 of these sources. Thus, about half of the sources classified as H II region are detected as compact 5 GHz radio continuum sources, presumably emitting free-free emission. The CORNISH source names are identified in Table 1.

Sources that were not detected by CORNISH may well be H II regions, but failed to be detected or cataloged by CORNISH. H II regions with large angular extent, for example, might be spatially filtered out by the VLA. Interferometric imaging of the Galactic plane often contains imaging artifacts and source extraction is sometimes difficult. Finally, some sources may lie below the CORNISH sensitivity limit. The

Spitzer images of sources identified as H II regions which were detected with CORNISH are not obviously distinguishable from those that were not. Consequently, we retain the IR-based classification, but caution the reader that the CORNISH non-detections may represent objects that are either more diffuse or produce less ionizing flux than the CORNISH detections.

3. RESULTS

The number of sources to be analyzed in the full sample of 981 sources (862 high-mass clumps and 119 IRDCs) was reduced to 334 after rejecting those with insufficient signal-to-noise, multiple velocity components, and poor fits. Of the remaining 334 sources, we find that 174 have significant ($\geq 3\sigma$ deviation from unity) outer-satellite HIA ratios ($\text{HIA}_{\text{outer}} < 1$), 76 have significant inner-satellite HIA ratios ($\text{HIA}_{\text{inner}} > 1$), and 44 have significant anomalies in both inner- and outer-satellite ratios. We do not find any sources that display a significant “inverse” outer HIA, specifically values of $\text{HIA}_{\text{outer}} > 1$. However, we do find five sources that exhibit a significant inverse inner HIA, with values of $\text{HIA}_{\text{inner}} < 1$. Derived rotational temperatures range from ~ 12 –25 K. For the full sample, we calculate an average $\text{HIA}_{\text{outer}}$ value of 0.812 ± 0.004 and an average $\text{HIA}_{\text{inner}}$ value of 1.125 ± 0.005 ; this yields deviations from unity of 47.5σ and 25σ respectively.

In the sample of IRDCs, the number of sources to be analyzed was reduced from 119 to 24, after again rejecting those with insufficient signal-to-noise, multiple velocity components, and poor fits. Of the remaining 24 sources, we find that 13 have significant ($\geq 3\sigma$ deviation from unity) outer-satellite HIA ratios ($\text{HIA}_{\text{outer}} < 1$), 3 have significant inner-satellite HIA ratios ($\text{HIA}_{\text{inner}} > 1$), and 1 has a significant anomaly in both outer- and inner- satellite ratios. We do not find any sources that display a significant inverse HIA ($\text{HIA}_{\text{outer}} > 1$ or $\text{HIA}_{\text{inner}} < 1$). Derived rotational temperatures range from ~ 13 –23 K. For this IRDC subsample, we calculate an average $\text{HIA}_{\text{outer}}$ value of 0.81 ± 0.02 and an average $\text{HIA}_{\text{inner}}$ value of 1.09 ± 0.02 , yielding deviations from unity of 9.5σ and 4.5σ respectively.

Along with evolutionary classification, we present a short extract of these results in Table 1.

4. DISCUSSION

4.1. The Ubiquity of the HIA

While there is not a significant detection of the HIA in every high-mass star-forming region observed, we find that a majority of sources in both samples exhibit the HIA: 62% of sources in the full sample and 63% of sources in the IRDC subsample show anomalies in either the inner- or outer-satellite HIA ratio. The ubiquity of the HIA is further evidenced by the HIA ratio sample averages, which deviate significantly from the LTE prediction of a unity ratio. Specifically, for the full sample the average HIA ratios for the inner ($\text{HIA}_{\text{inner}}$) and outer ($\text{HIA}_{\text{outer}}$) satellites deviate from unity by 19% (47.5σ) and 13% (25σ) respectively. For the IRDC subsample the average inner and outer HIA ratios also significantly deviate from unity, with the average outer ratio ($\text{HIA}_{\text{outer}}$) deviating by 19% (9.5σ) and average inner ratio ($\text{HIA}_{\text{inner}}$) deviating by 9% (4.5σ). This can also be seen by plotting I_{LO} versus I_{RO} and I_{LI} versus I_{RI} (Figure 2). From this plot, we observe that the samples of HIAs follow average anomalies (solid blue lines) that clearly differ from unity (dashed red lines). In high-mass star-forming

Table 1
Source Properties and HIA Results

Source Name	Coordinates		T_{rot} (K)	HIA ^a (outer)	HIA ^b (inner)	Classification ^c	Optical Depth (τ)
	l°	b°					
G5.39+0.19	5.389	0.188	15.2 ± 1.8	0.87 ± 0.14	1.11 ± 0.17	Unclassified	1.85 ± 0.26
G5.62-0.08	5.618	-0.082	16.4 ± 0.7	0.95 ± 0.05	1.09 ± 0.05	Protostellar	2.31 ± 0.09
G5.64+0.24	5.638	0.237	18.7 ± 0.9	0.69 ± 0.04	1.15 ± 0.06	H II	1.82 ± 0.10
G5.83-0.51	5.832	-0.514	15.3 ± 0.6	0.82 ± 0.05	1.04 ± 0.05	Unclassified	1.49 ± 0.08
G5.89-0.29	5.893	-0.287	15.9 ± 1.1	0.61 ± 0.07	1.13 ± 0.10	Protostellar	1.53 ± 0.15
G5.89-0.32	5.893	-0.318	19.0 ± 1.3	0.74 ± 0.08	0.86 ± 0.08	Unclassified	1.17 ± 0.14
G5.90-0.43	5.900	-0.427	22.6 ± 0.9	0.78 ± 0.04	1.14 ± 0.05	Unclassified	1.68 ± 0.07
G5.90-0.44	5.898	-0.443	23.5 ± 1.4	0.55 ± 0.06	0.80 ± 0.07	Unclassified	1.19 ± 0.13
G6.13-0.63	6.125	-0.629	17.4 ± 1.7	0.72 ± 0.10	1.04 ± 0.13	Protostellar	1.48 ± 0.19
G6.19-0.36	6.189	-0.356	19.8 ± 0.9	0.73 ± 0.04	1.24 ± 0.06	Quiescent	1.85 ± 0.09

Notes.

^a Defined as NH_3 (J, K) = (1, 1) ratio of outer satellite intensities $I_{\text{LO}}/I_{\text{RO}}$.

^b Defined as NH_3 (J, K) = (1, 1) ratio of inner satellite intensities $I_{\text{LI}}/I_{\text{RI}}$.

^c Classification based on Chambers et al. (2009) using *Spitzer* IR data.

^d Denotes a source detected by the CORNISH survey: cornish.leeds.ac.uk

(This table is available in its entirety in machine-readable form.)

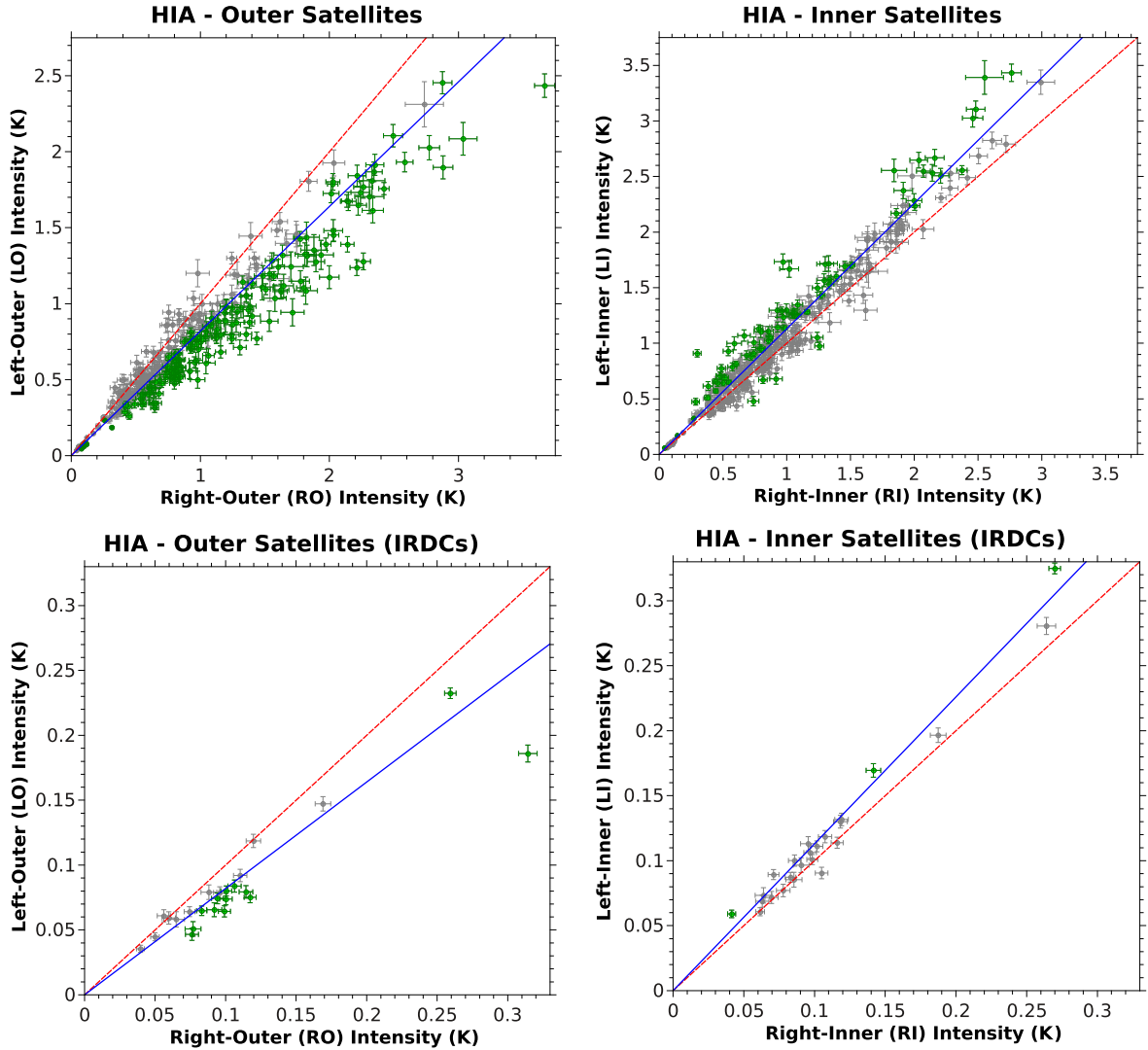


Figure 2. Plots showing the deviation from unity of the (J, K) = (1, 1) inner- and outer-satellite ratios for both the full sample (upper) and the subsample of IRDCs (lower). The dashed red line is unity and the blue line represents the sample average ratio. In LTE, the source points are expected to lie along unity. Left: plots of $F = 1 \rightarrow 0$ “left-outer” (LO) satellite intensity vs. the $F = 0 \rightarrow 1$ “right-outer” (RO) satellite intensity for the full sample (upper left) and IRDCs (lower left). Right: plots of $F = 1 \rightarrow 2$ “left-inner” (LI) satellite intensity vs. the $F = 2 \rightarrow 1$ “right-inner” (RI) satellite intensity for the full sample (upper right) and IRDCs (lower right). The green points represent those sources with significant ($>3\sigma$ deviation from unity) HIA ratios.

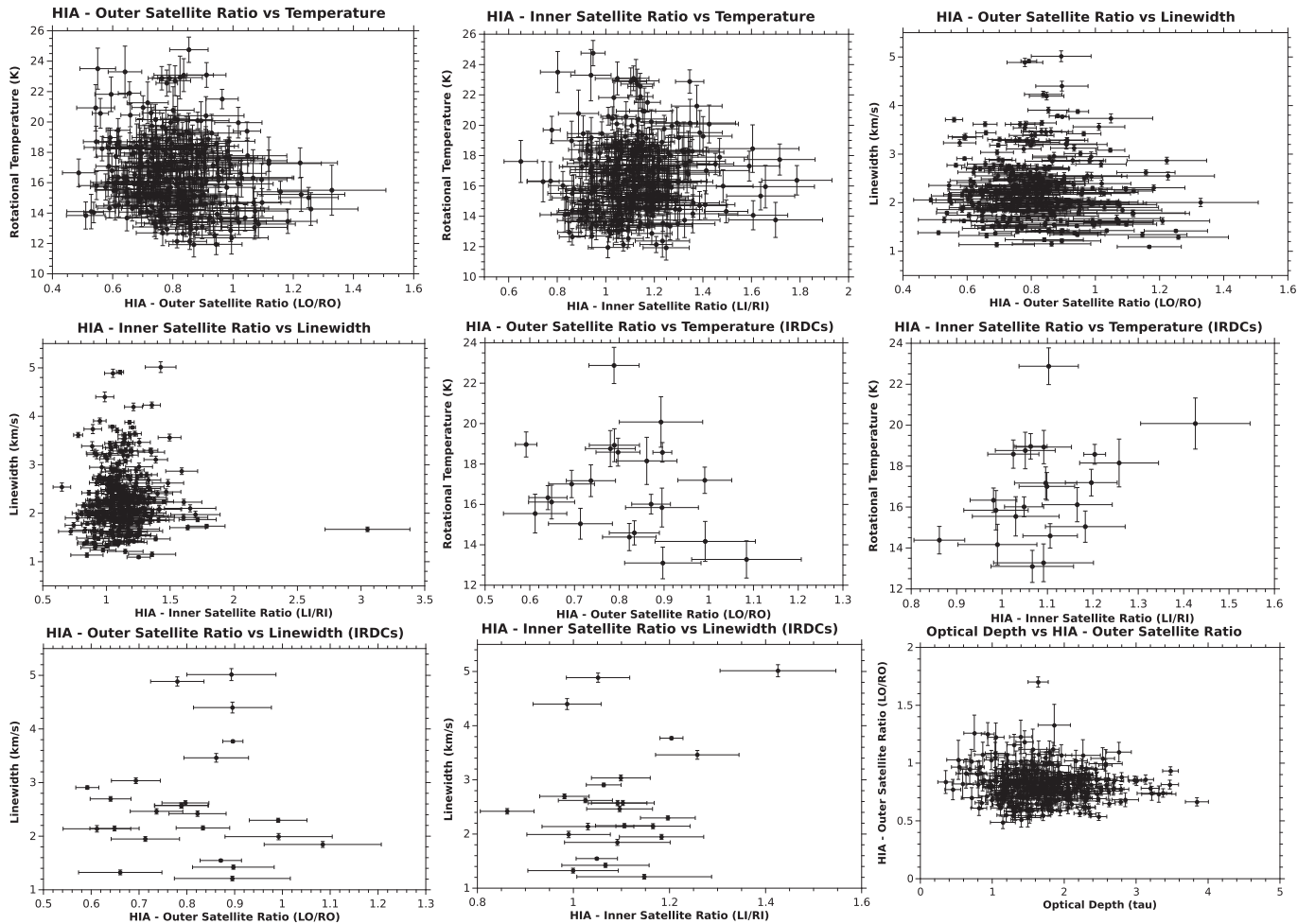


Figure 3. Plots of the HIA vs. rotational temperature, line width, evolutionary category, and optical depth for the full sample and IRDC subsample.

regions NH_3 spectra exhibiting asymmetric satellite intensities are more common than those exhibiting symmetric, LTE-predicted satellite intensities.

The ubiquity of the HIA suggests a mechanism that reflects a common process or condition in star-forming regions. The sample clearly shows that $\text{HIA}_{\text{outer}} < 1$ and $\text{HIA}_{\text{inner}} > 1$. The deviation from unity for these anomalies is predicted by the HIA “core” model of Stutzki & Winnewisser (1985). However, the results are inconsistent with the systematic collapse or expansion models of Park (2001), in which the $\text{HIA}_{\text{outer}}$ and $\text{HIA}_{\text{inner}}$ values deviate from unity in the same sense. In the Park models, expanding motions give rise to an enhancement of both of the “left” hyperfines (both $\text{HIA}_{\text{outer}}$ and $\text{HIA}_{\text{inner}}$ are < 1) and collapsing motions gives rise to an enhancement of both of the “right” hyperfines (both $\text{HIA}_{\text{outer}}$ and $\text{HIA}_{\text{inner}}$ are > 1). The Park (2001) expansion and collapse models can have opposite senses of the deviation from unity for $\text{HIA}_{\text{outer}}$ and $\text{HIA}_{\text{inner}}$, but only if the HIA is dominated by the core model mechanism of Stutzki & Winnewisser (1985), and the systemic motions are very small. The results, therefore, suggest that in high-mass star-forming regions and IRDCs, the HIA core model best accounts for the observed HIAs. This suggests that these regions are highly clumped on small scales.

The clumpy structure of star-forming regions has been inferred for decades (see review by Wilson & Walmsley 1989), and thus it may not be surprising that HIAs that require

clumping are found in these high-mass star-forming regions and IRDCs. However, the HIA core model mechanism specifically requires cores with very narrow line widths ($\sim 0.3 \text{ km s}^{-1}$), much smaller than the observed line widths ($\sim 3 \text{ km s}^{-1}$). For virialized cores, the inferred core masses are $\sim 1 M_{\odot}$. Thus, if the HIA core model mechanism is indeed responsible for the observed ubiquity of HIAs, at small scales the structure of molecular clumps should be dominated by cores with masses $\sim 1 M_{\odot}$.

A ubiquitous HIA, therefore, may have implications for theories of high-mass star formation, specifically the “competitive accretion” model. In the competitive accretion model, high-mass stars form from massive clumps ($\sim 100\text{--}1000 M_{\odot}$) that in their earliest phases fragment into a large number of dense molecular cores with masses near the Jeans mass, or $\sim 1 M_{\odot}$ (Bonnell et al. 2001; Zinnecker & Yorke 2007). The cores then “compete” to acquire the remaining material in the clump, with those closest to the center (i.e., deeper in the potential well) eventually accreting enough mass to become high-mass stars. These initial core fragments are predicted to have temperatures of $\sim 10\text{--}20 \text{ K}$, which yield thermal line widths $\lesssim 0.3 \text{ km s}^{-1}$ and core diameters of $\sim 0.05 \text{ pc}$ at typical densities of $\sim 10^5 \text{ cm}^{-3}$. Such initial conditions are in good agreement with the HIA core model. Therefore, if the HIA core model is correct, the HIA may provide indirect evidence and support for competitive accretion: a signpost indicating that in

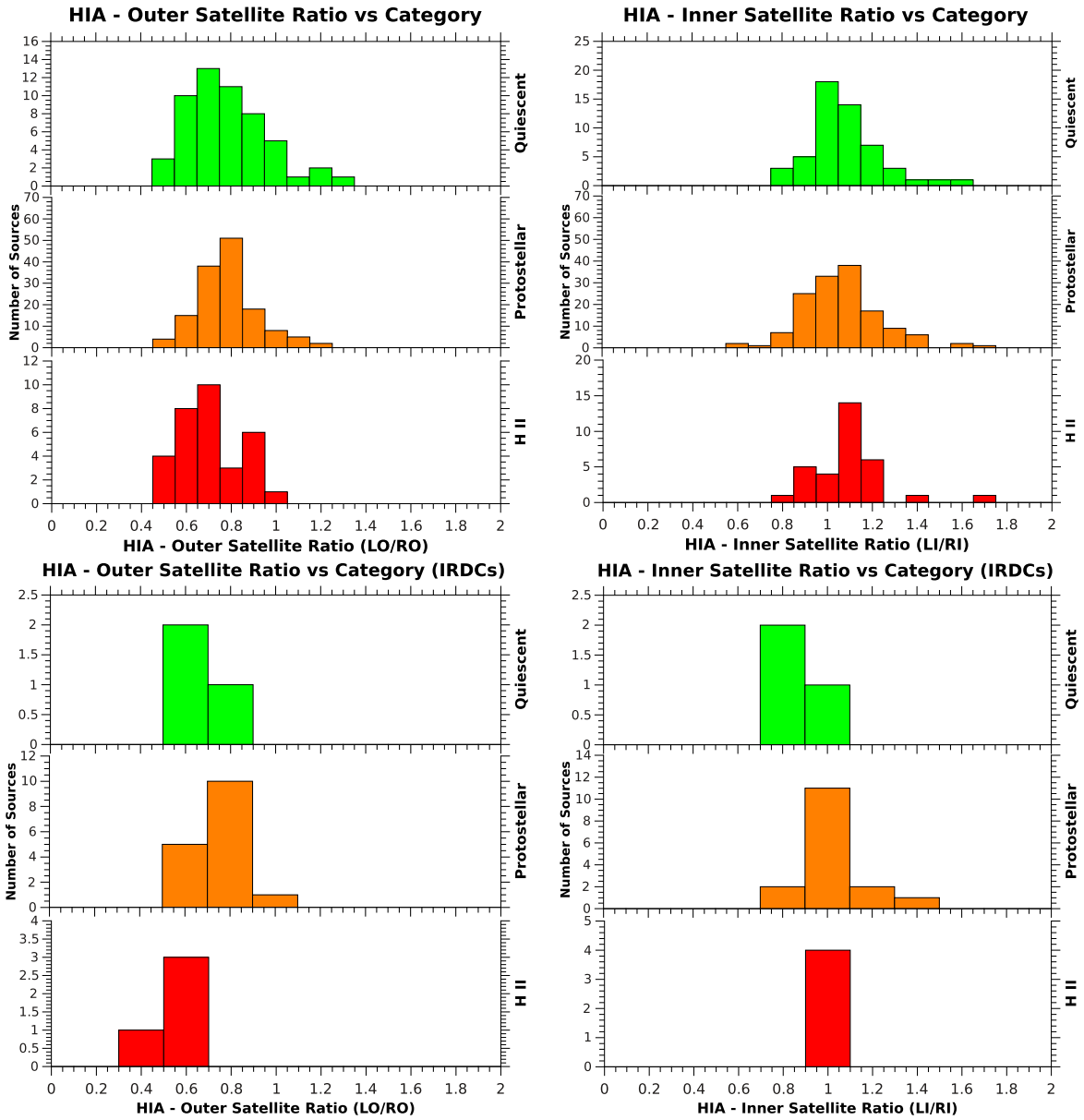


Figure 4. Distributions of HIA ratios (inner and outer) vs. evolutionary classification category for both the full sample and IRDC subsample.

the early stages of high-mass star formation $\sim 1 M_{\odot}$ cores with the required narrow line widths predominate.

Despite the ability of the HIA core model to account for the observed HIAs and its apparent consistency with competitive accretion, the HIA core model implicitly assumes that the small-scale structure of molecular clumps consists of a uniform population of virialized cores with similar properties. This is surely an oversimplification, as real clumps are turbulent and are known to have structure at all spatial scales. Indeed, recent work (e.g., Kainulainen et al. 2009) suggests that the column density probability distribution or N-PDF of molecular clumps is typically characterized by a log-normal distribution with a power-law tail at high-column densities. This distribution is inconsistent with a clump consisting solely of uniform cores at small scales. For clumps whose internal structures are dominated by turbulence, the turbulent cascade naturally leads to a scaling between velocity dispersion and spatial scale. For a sample of giant molecular clouds, Roman-Duval et al. (2011)

find that a velocity fluctuation of 0.3 km s^{-1} corresponds to a spatial scale of $\sim 0.3 \text{ pc}$. Thus, the Stutzki & Winnewisser (1985) model might also produce HIAs from a turbulent medium if the NH_3 emission is dominated by structures with spatial scales of 0.3 pc or smaller. To test whether a clump whose structure is dominated by turbulence is capable of producing HIAs would require a detailed radiative transfer model of the hyperfine line emission through a turbulent medium.

Nevertheless, the fact that HIAs are ubiquitous in high-mass star forming regions and in IRDCs points to a commonplace condition. The evidence suggests that small scale gas structures with narrow line widths, generated by fragmentation into cores with masses $\sim 1 M_{\odot}$ or by turbulence, are required to produce the observed HIAs. It remains unclear, however, how to reconcile the observed large line widths of $\sim 3 \text{ km s}^{-1}$ with the requirement for narrow line widths of $\sim 0.3 \text{ km s}^{-1}$ on small scales.

Table 2
HIA Correlation Results

Property 1	Property 2	Satellites ^a	Sample	R^b	K-S ^c
HIA ratio (LO/RO)	Temperature	Outer	Full	-0.21	...
HIA ratio (LI/RI)	Temperature	Inner	Full	0.21	...
HIA ratio (LO/RO)	Line width	Outer	Full	-0.22	...
HIA ratio (LI/RI)	Line width	Inner	Full	0.20	...
HIA ratio (LO/RO)	Optical depth	Outer	Full	-0.30	...
Quiescent HIA ratios (distribution)	Protostellar HIA ratios (distribution)	Outer	Full	...	0.49
Quiescent HIA ratios (distribution)	H II HIA ratios (distribution)	Outer	Full	...	0.29
Protostellar HIA ratios (distribution)	H II HIA ratios (distribution)	Outer	Full	...	0.02
Quiescent HIA ratios (distribution)	Protostellar HIA ratios (distribution)	Inner	Full	...	0.50
Quiescent HIA ratios (distribution)	H II HIA ratios (distribution)	Inner	Full	...	0.53
Protostellar HIA ratios (distribution)	H II HIA ratios (distribution)	Inner	Full	...	0.29
HIA ratio (LO/RO)	Temperature	Outer	IRDC	-0.22	...
HIA ratio (LI/RI)	Temperature	Inner	IRDC	0.37	...
HIA ratio (LO/RO)	Line width	Outer	IRDC	-0.31	...
HIA ratio (LI/RI)	Line width	Inner	IRDC	0.46	...
Quiescent HIA ratios (distribution)	Protostellar HIA ratios (distribution)	Outer	IRDC	...	...
Quiescent HIA ratios (distribution)	H II HIA ratios (distribution)	Outer	IRDC	...	...
Protostellar HIA ratios (distribution)	H II HIA ratios (distribution)	Outer	IRDC	...	...
Quiescent HIA ratios (distribution)	Protostellar HIA ratios (distribution)	Inner	IRDC	...	...
Quiescent HIA ratios (distribution)	H II HIA ratios (distribution)	Inner	IRDC	...	...
Protostellar HIA ratios (distribution)	H II HIA ratios (distribution)	Inner	IRDC	...	...

Notes.

^a Denotes whether the statistic applies to the NH₃ (1, 1) inner or outer satellites.

^b Correlation coefficient.

^c Two-sample K-S test statistic. Defined as the probability that the two distributions are drawn from the same parent distribution.

4.2. Correlations Between HIAs and High-mass Star-forming Region Properties

If the HIA core model accurately describes the mechanism giving rise to the HIA, and in fact traces the earliest stages of high-mass star formation, then the strength of the HIA should vary with clump and core properties that relate to clump evolution. Because the HIA core model requires line widths of $\sim 0.3 \text{ km s}^{-1}$, we expect the strength of the anomaly to decrease (HIA ratio to move closer to unity) with increasing temperature or line width (e.g., turbulence), and in turn correlate with evolutionary stage. In other words, quiescent (prestellar) high-mass star-forming regions with lower temperatures should be more capable of hosting the conditions required for narrow line widths than hotter and more turbulent regions with embedded protostars, hot molecular cores, outflows, and/or H II regions.

Temperatures in regions in more active stages of high-mass star formation are typically higher than those in the quiescent stage (Chambers et al. 2009; Hoq et al. 2013). IRDCs, hosting many cold, quiescent clumps, typically show temperatures $< 25 \text{ K}$ (Rathborne et al. 2006, 2010). Molecular transitions often observed in protostellar and H II regions indicate temperatures near or above $\sim 100 \text{ K}$ (Kurtz et al. 2000). In addition, observations of NH₃ in protostellar cores in IRAS 18566+0408 yield rotational temperatures of 45 K , with higher temperatures up to 70 K in associated outflows (Zhang et al. 2007).

To test the hypothesis that the strength of the HIA is most extreme in the earliest phases of high-mass protostellar evolution, we compare the HIA ratios of individual sources to their temperature, line width, and evolutionary classification (see Section 4.3 for discussion of optical depth). Again, we expect HIA ratios to be inversely related to temperature and

line width. We also expect HIA ratios to evolve, with the most significant HIAs in the quiescent phase, and then approaching the LTE value of unity in the protostellar and H II region phases. However, we do not find any significant correlation between either inner or outer HIA ratios and temperature or line width (Figure 3). We also do not find any significant trend in either sample between HIA strength and evolutionary classification (Figure 4). In order to determine whether the sources classified as quiescent, protostellar, or H II region belong to distinct populations, we perform a two-sample Kolmogorov-Smirnov (K-S) test for each pair of distributions (separately for the inner- and outer-satellite HIA ratios) and determine the probability that each is drawn from the same parent distribution. This is done only for the full sample, since the IRDC sample contains too few sources to perform the K-S test. We find that the data are consistent with all of the distributions being drawn from the same parent distribution, though there is a high probability (98%) that the outer-satellite protostellar distribution is distinct from the outer-satellite H II region distribution. Given this, and the significant overlap among all of the distributions, there is insufficient evidence to establish a trend in HIA ratio with evolutionary classification. We report these K-S test statistics, along with the correlation coefficients for HIA ratio versus temperature and line width, in Table 2.

The lack of any trend in the strength of the HIA contradicts the simple prediction that the NH₃ emission is dominated at early stages by a population of narrow line width cores that evolve into sources in which the emission is dominated by warmer, higher-mass, broad line-width cores. There are several possible reasons for this discrepancy. First, it is possible that the proposed HIA core model may be incorrect. Although this model is in broad agreement with the observations, it may

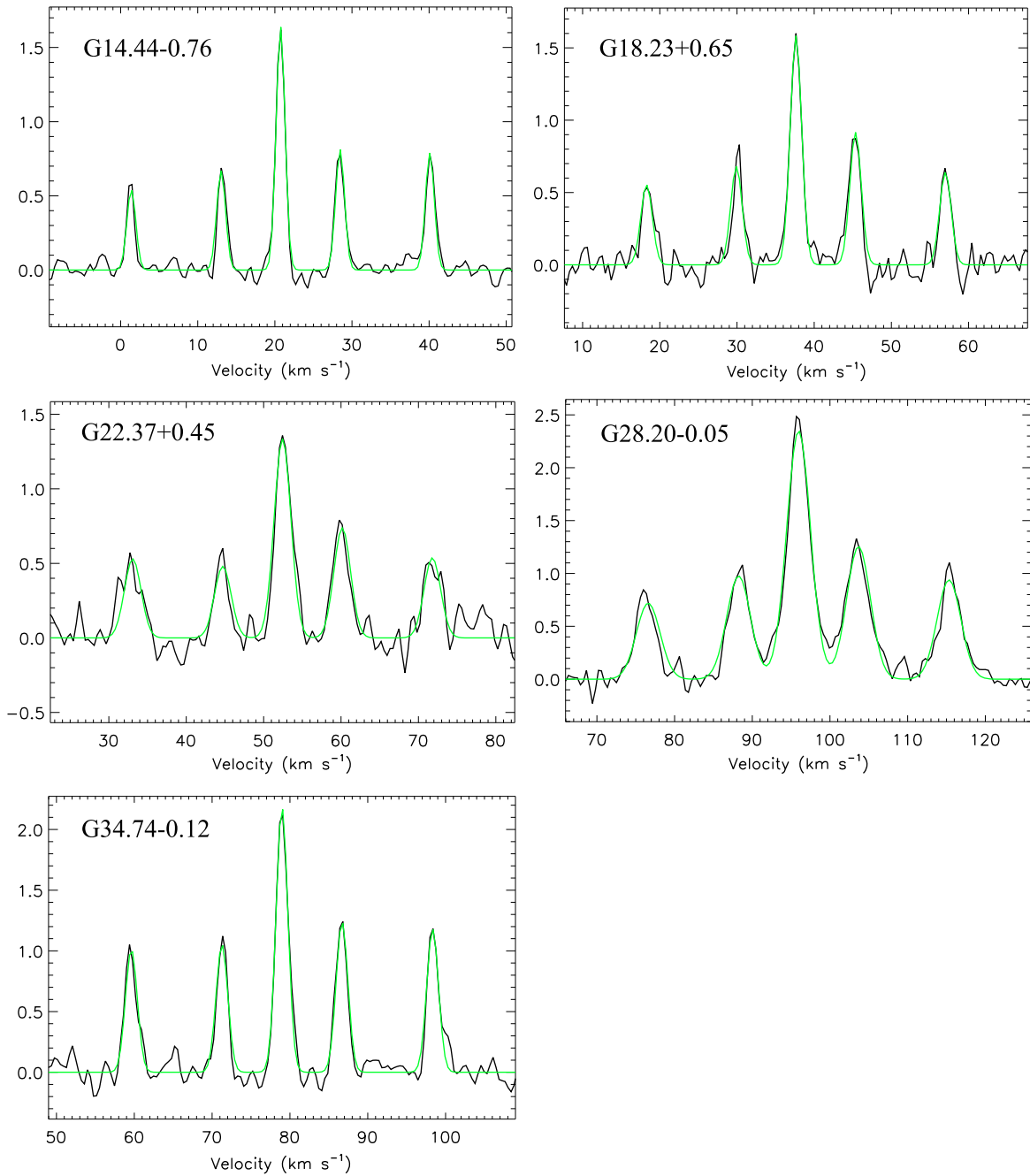


Figure 5. Spectra of five sources showing significant inverse inner HIAs, namely $I_{LI}/I_{RI} < 1$. These spectra are characterized by values of $I_{RI} > I_{LI}$ and $I_{RO} > I_{LO}$.

oversimplify the velocity and density structure of actual star-forming regions. That is, the suggestion that the NH_3 emission arises solely from an ensemble of small cores is highly idealized and oversimplified. For instance, in considering a fully turbulent gas cloud in the absence of cores, there may be lines of sight with narrow line widths that arise by chance due to turbulence. In this case, the gas is not organized into a large number of narrow line-width cores, and the anomaly could arise from unrelated gas pockets along the line of sight. Thus, if the NH_3 emission is dominated by turbulent gas, and not cores, no HIA evolution would result. Moreover, the simple evolutionary scheme, in which the early stages are dominated by narrow-line width cores and later stages by broad line-width cores, may not reflect the actual evolution of high-mass clumps. Indeed, in the turbulent core accretion model of high-mass star

formation, one might expect broad line-width, high-mass cores in the early stages (McKee & Tan 2003). Even in the competitive accretion model, it is not clear that clumps in the quiescent phase are sufficiently young to harbor only narrow line width cores.

Another possibility is that the angular resolution of Effelsberg ($\sim 40''$) and Parkes ($\sim 50''$) is insufficient to distinguish the evolution of the HIA. In these relatively large beams, the observed NH_3 emission may include the blended emission from narrow line-width cores (with strong HIAs) and broader line-width gas, either from diffuse inter-core gas or an admixture of warmer, more massive cores. In this case, the evolution of cores may be difficult to observe with a large telescope beam due to the blending of multiple gas components.

4.3. Optical Depths

In LTE, the optical depth of the NH_3 (1, 1) line can be determined from the ratio of the intensities of the main hyperfine line to the satellite hyperfine lines. If the lines are optically thin these ratios will equal the relative statistical weights, and as the optical depth increases the intensity ratio approaches unity. The proper determination of optical depths in the presence of HIAs, however, is not so obvious, since non-LTE excitation, in addition to optical depth, alters the hyperfine intensity ratios. (See Section 2 for details on the methodology used to calculate optical depths and their associated uncertainties.)

Nonetheless, in the HIA models, optical depth plays an important role because of selective trapping, where some of the hyperfine lines have a larger escape probability than others. Thus, one might expect the HIA values to depend on optical depth, with stronger HIA departures from unity at higher optical depths. Figure 3 (bottom-right panel) displays a plot of the optical depth versus the $\text{HIA}_{\text{outer}}$ value. The figure shows marginal evidence for HIAs closer to unity at the smallest optical depths, but the trend is weak at best, with a correlation coefficient of -0.30 (Table 2). It is puzzling that the trend is not stronger. A number of issues might contribute. For example, the derivation of optical depth in the presence of HIAs, or the blending of large and small scale structures within the single-dish telescope beam, may both be problematic.

4.4. Inverse HIAs

The HIA core model accounts for asymmetric NH_3 (1, 1) satellite line profiles with only one pattern: $I_{\text{LO}} < I_{\text{RO}}$ and/or $I_{\text{LI}} > I_{\text{RI}}$. Though the majority of our sources exhibit this line intensity profile, five sources show significant inverse inner anomalies, with $I_{\text{LI}} < I_{\text{RI}}$ (Figure 5). Simulations by Park (2001), adapting the original HIA core model calculations of Stutzki & Winnewisser (1985) to include systematic core expansion or contraction, can account for profiles with $I_{\text{LO}} > I_{\text{RO}}$ or $I_{\text{LI}} < I_{\text{RI}}$. In systematic expansion, the inner HIA is inverted, while in systematic contraction the outer HIA is inverted. Thus, in the case of systematic core expansion both $I_{\text{RO}} > I_{\text{LO}}$ and $I_{\text{RI}} > I_{\text{LI}}$. Conversely, in the case of systematic core contraction, $I_{\text{RO}} < I_{\text{LO}}$ and $I_{\text{RI}} < I_{\text{LI}}$.

The sources G14.44–0.76, G28.20–0.05, and G34.74–0.12 each show a significant core-expansion profile, with well-separated inner- and outer-satellite line amplitudes. That is, the fitted line amplitudes, including uncertainties, yield separations for $I_{\text{RO}} > I_{\text{LO}}$ and $I_{\text{RI}} > I_{\text{LI}}$ of $>3\sigma$. For the remaining two sources G22.37+0.45 and G18.23+0.65 the fitted line amplitudes of the inner satellites ($I_{\text{RO}} > I_{\text{LO}}$) differ by $>3\sigma$, but those of the outer satellites ($I_{\text{RI}} > I_{\text{LI}}$) differ by $<2\sigma$. Therefore, the spectra of the five sources exhibiting inverse inner HIAs are consistent with cores undergoing systematic expansion, with intensity profiles showing $I_{\text{RO}} > I_{\text{LO}}$ and $I_{\text{RI}} > I_{\text{LI}}$.

5. SUMMARY AND CONCLUSIONS

We have analyzed the NH_3 HIA toward 334 high-mass star-forming regions (310 massive ($\gtrsim 100 M_\odot$) clumps and 24 IRDCs). $\text{NH}_3(J, K) = (1, 1)$, $(2, 2)$, and $(3, 3)$ spectral line data were obtained from publicly available Effelsberg Telescope observations and our own observations with the Parkes Telescope. For each source, we have calculated inner and outer HIA ratios, rotational temperatures, and optical depths. In

addition, we have compared the HIA to NH_3 rotational temperature, spectral line width, optical depth and evolutionary classification. From this, we draw the following conclusions:

1. The HIA is ubiquitous among high-mass star-forming regions. The majority of sources exhibit significant ($>3\sigma$ deviations from unity) HIA ratios, with 62% of sources in the full sample and 63% of sources in the IRDC subsample showing anomalies either in the inner- or outer-satellite HIA. Average HIA ratios in each sample deviate from unity with levels of significance from 4.5σ – 47.5σ . If the HIA core model is correct, this may suggest that $\sim 1 M_\odot$ cores predominate the early stages of high-mass star formation, a feature consistent with competitive accretion.
2. No significant correlation exists between the HIA and temperature, line width, or optical depth. Additionally, no significant trend exists between the HIA and stage of high-mass stellar evolution. This is curious, as we expect the HIA to be inversely related to increasing temperature and line width, due to the requisite narrow $\sim 0.3 \text{ km s}^{-1}$ line widths. This may indicate that HIAs do not result from gas structures dominated by evolving, virialized cores (perhaps instead dominated by turbulence), or that HIA evolution may not be distinguishable with large beam sizes due to the blending of emission from multiple gas components.
3. Though the majority of sources exhibit the HIA predicted by the HIA core model ($I_{\text{LO}} < I_{\text{RO}}$ and/or $I_{\text{LI}} > I_{\text{RI}}$), a few sources exhibit asymmetries consistent with cores undergoing systematic expansion. This causes the inner HIA to invert from $I_{\text{LI}} > I_{\text{RI}}$ to $I_{\text{LI}} < I_{\text{RI}}$.

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